



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN
(INFORMATION SHEET)
JABATAN PEMBANGUNAN KEMAHIRAN

KOD DAN NAMA PROGRAM / <i>PROGRAM CODE & NAME</i>	Z-009-4:2015 CORE ABILITIES	
TAHAP / LEVEL	4 (FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	C02:HEALTH, SAFETY AND ENVIRONMENTAL MONITORING	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	2.1 ENSURE HEALTH AWARENESS PROGRAM 2.2 ENSURE SAFETY AWARENESS PROGRAM 2.3 ENSURE ENVIRONMENTAL AWARENESS PROGRAM	
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TAJUK/TITLE: SAFETY AND HEALTH AWARENESS PROGRAMME

TUJUAN/PURPOSE:

This information sheet is intended to describe on condition and aspects to consider in the development of a safety and health programme and safety Awareness Programme at Workplace. The person who is competent in this CA shall be able to ensure safety and health awareness programme at workplace.

PENERANGAN/INFORMATION:

1.0 INTRODUCTION

While organisation strive towards workplace safety and health, accidents continue to occur. Having a safety and health programme in place is not a guarantee that accidents will not happen. Safety and health is the responsibility of all, from the management level to the

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supervisors and the employees. Everyone must play their role and contribute to the successful implementation of the programme. Other factor affecting the successful implementation of a safety and health programme are :

- Management commitment on the allocation of monetary budget and the demonstration of job safety and health;
- Training of supervisor and new employees;
- An established written safety and health programme;
- Accountability for safety and health at all levels;
- Response and compliance to accident investigation; and
- Conduct safety and health inspection and audit

2.0 DEVELOPING A SAFETY AND HEALTH PROGRAMME

Safety and health programme, must be incorporated into business activities to ensure that all aspect of safety and health management are covered. Three condition for a successful safety and health programme :

- i. Management leader ship
- li. Safe Working Condition
- lii. Safe working habits by all employees

The objectives of a safety and health programme are ensure ;

- The company has a direction and commitment for safety and health that include policies and procedures for all employees and other interested parties.
- All misunderstanding and misconceptions are clarified for a specific safety and health activity or issue;
- The practise of safety and health at the workplace is part of the objectives the performance measurement;
- There is a written safety and health rules and procedure for every to follow;
- That all activities have been integrated with safety and health requirement;
- Proper communication tools are established for safety and health communication

With the effective implementation of a written safety and health programme, it is expected to reduce unwanted incident or accidents and also damage to property.

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3.0 FACTORS DETERMINING THE SUCCESS OF A SAFETY AND HEALTH PROGRAMME

Factors that determine the success of safety and health programme :

- Management Commitment and Employee Involvement
- Worksite Analysis
- Hazard Prevention and Control
- Safety and Health Training

3.1 Management Commitment and Employee Involvement

To ensure that safety and health programme can be continuously evaluated, the management must establish a clear goal and objective for the safety and health programme. Management have to provide visibility and involvement in implementing the safety and health programme through :

- Commitment to safety and health policy
- Establishment of safety and health consultative committee
- Involvement in providing resources for safety and health programme.

3.2 Worksite Analysis

Worksite analysis is an activity that should be included in the safety and health programme which is to identify hazard and potential hazards that may exist in work processes and in the working environment. The following are performed to identify workplace hazards or potential hazards :

- Conduct comprehensive periodical surveys on hazards identification
- Analyse new processes, methods, materials, equipment and installation
- Conduct Hazard Identification, Risk Assessment and Risk Control (HIRARC) or job hazards analysis
- Conduct regular site safety and health inspection
- Provide a system that is reprisal –free for employees that notify management of hazardous conditions

3.3 Hazards Prevention and Control (Risk Management)

Occupational safety and health legislation requires anyone in control of the workplace to identify the potential hazards of the proposed work, assess the risks involved and develop controls to eliminate or minimize the risk. Once hazards or potential hazards are identified, the management has to establish procedures to prevent exposure to

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employees and to control the hazards. The Procedures shall include the hierarchy of control to ensure that the priority for hazards prevention is followed accordingly which include :

- Elimination
- Substitution
- Isolation
- Engineering Control
- Administrative Control
- Personal Protective Equipment (PPE)

3.4 Safety and Health Training

Training and education helps to ensure that employees are adequately trained to perform their safety and to raise their awareness on workplace and health.

An effective safety and health training addresses the following:

- The responsibility of everyone in the organisation with regards to safety and health programmes .
- Integrates safety and health training into other training
- Defines the complexity of the safety and health training, depending on the size and complexity of the workplace and the natureof hazards that employees are exposed to.

4.0 SAFETY AWARENESS PROGRAM IN THE WORK PLACE

4.1 Occupational Safety and Health Management System (OSHMS) In The Organisation

The system shall contain the main elements categories are:

- Policy
- Organising
- Planning & Implementing
- Evaluation
- Action for Improvement

4.2 Safety and Health Audit

The safety and health audit is important in ensuring the safety of a plant environment, workplace or equipment and that the people are working safely. Audits are normally performed for the following reasons ;

- To identify hazards
- To ensure compliance with company rules and regulation
- To ensure compliance with the relevant authority's requirements
- To determine the safety and health condition of a workplace.
- To determine the safety and health condition of equipment and machineries.
- To determine the safety and health condition of operation processes
- To evaluate the performance of safety and health management.
- To evaluate supervisor awareness towards safety and health.
- To evaluate employees performance in regards to safety and health.
- To evaluate the progress and effectiveness of safety and health issues and problem.
- To determine the effectiveness of hazards control at a new workplace, new equipment or new processes.

4.3 Emergency Prevention, Preparedness and Response

Emergency prevention, preparedness and response is important to ensure that company is ready to face any unexpected emergency situation. Important aspects to consider in your emergency prevention, preparedness and response is summarised in table below.

Aspect	Description
Emergency planning and preparation	Includes employee preparation, response and evacuation during an emergency – train employees to take the appropriate actions during an emergency, such as organising evacuation drills to familiarise them with the emergency evacuation routes. The management must also establish an emergency response team to ensure there are specially trained personnel in the company to handle emergency situations.
Emergency Equipment	Ensure that the organisation has all the necessary equipment and tool to handle potential emergencies appropriately, the equipment and tools must be tested, serviced, calibrated and maintained periodically to ensure their functionality.
Medical	The medical team also has to be always prepared to go on site

Programme	to attend to employees for emergency care, the medic personnel must be available on site with appropriate medication and equipment.
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4.4 Promotions

The promotion of occupational safety and health, as part of an overall improvement of working conditions, represent an important strateg. Promotions program could be in the form of safety week, poster & slogan Competiitions, safety award schemes, photographic displays, regular site safety tours, exhibitions of safety equipment and short safety talks, film shows etc.

4.5 Tool Box Talk

A tool box talk is a hands on way to remind workers that safety and health are important on the job which deal with specific problems on site. Through tool box talks we can tell workers about safety and health requirements for the tool, equipment, material and procedures they use every day or for particular jobs. The objective of tool box is to help workers recognize and control hazards on the project which demonstrate the commitment of employers and workers to safety and health on the job.

4.6 Permit To Work System

An area where work has to be undertaken in a particularly hazardous situation or environment, such as working on or near existing line and in confined spaces e.g. tunnel and high voltage room, or where workers can be jeopardized by unwitting action of other, or by their own failure to realize an actual or potential danger etc.

- i) Types Of Permit Used in Work Activities
 - a. Welding / burning (naked flame) work permit
 - b. Hot work (no naked flame) work permit.
 - c. Flammable release work permit.
 - d. Vessel entry work permit (confined spaces)
 - e. Cold work permit.
 - f. Electrical work permit.
 - g. Safe excavation work permit
 - h. Working at heigh work permit

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4.7 Lock-Out Tag-Out System

A Lock-out tag-out system is a written practice which when followed, will allow workers to carry out their job function without undue risks. The system's procedure can be quite simple, yet precise, such as locking out and tagging out simple piece of machinery or complex, such as the locking out of entire process. A lock-out tag out may applied to electrical and machanical equipment in nature,

4.8 Workplace Inspection.

Safety& health inspection and audits often have negative implication, associated with faultfinding. A positive approach based on fact finding will produce better results, and cooperation from all those taking part in the process is workplace Inspection. The objectives workplace inspections are:

- To identify hazardous conditions and start the corrective process.
- To improve operation and condition.
 - i) There are a number of types of Inspection :
 - a. Statutory - Compliance with relevant legislative requirements
 - b. External - By enforcement Officials, insurers, consultants
 - c. Executive - Senior management tour
 - d. Scheduled - Planned at appropriate intervals by supervisor.
 - e. Introductory - Check on new or reconditioned
 - f. Continuous - By employee, supervisor, which can be formal and pre-planned, or informal

4.9 Preventitive Maintenance Programme (PMP)

The preventitive maintenance programme (PMP) is normally scheduled to take place on a daily, weekly, montly or yearly basis and use inspection checklists preventitive maintenance programme (PMP) can be considered as an accident prevention tool, as equipment failures are minimised, therefore reducing the potentially unsafe conditions. Good preventitive maintenance programme lead to ;

- Improved equipment efficiency
- Improved attitudes toward safety
- Higher degree of safety ownership ; and
- Lower Chances of incidents from happening.

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4.10 Accident Investigation and Reporting Technique

An accident is an unexpected or unplanned event in a sequence of events that can occur through a combination of causes, resulting due to;

- Physical injury, illness or disease to an individual;
- Damage to property; or
- A combination of the above.

An accident investigation is important as it helps organisations to continually improve workplace safety. The purpose of an accident investigation are;

- Prevent the recurrence of a same or similar accident;
- Continuous monitoring and measuring of the organisation's safety and health performance;
- Find ways or measure to prevent recurrences
- Find methods to determine the root cause of the accident and;
- Provide a structured method of investigating accident.

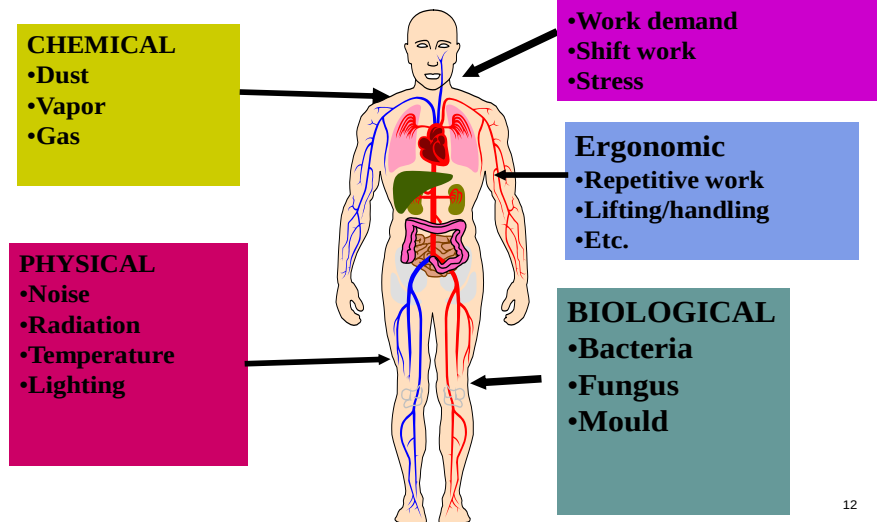
Accident prevention programme indirectly contributes to higher productivity and quality, as work processes are better organised and subsequently free from unnecessary interruptions. Accident prevention measures indirectly results in costly corrective measures when accident occur. There are four(4) techniques or tools for preventing accidents at the workplace;

- i) Job Hazard Analysis(JSA)
- ii) Job Safety Analysis(JSO)
- iii) Prevention Maintenance Programme(PMP)
- iv) Special Emphasis Programme

5.0 HEALTH AWARENESS PROGRAM IN THE WORK PLACE

Workplace health and wellness programs should be a part of the overall strategy for a healthy workplace. The purpose of a workplace health and wellness program is to offer a comprehensive health service for all employees. Health hazards to be included in the Health Awareness programme are;

Health Hazards at workplace



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5.1 Chemical Control Programme

Working with chemicals can involve the risk of exposure, becoming hazardous to a person's health. Those health risks are dependent upon the toxicity of the chemical, the types of effects, and how the chemicals enter the body. There are four major routes of entry chemical into the body:

- Inhalation (breathing)
- Absorption (skin contact)
- Ingestion (eating)
- Injection

The most common way workplace chemicals enter the body is by breathing. Other chemicals can be absorbed through the skin into the bloodstream. They can also be accidentally swallowed if hands or cigarettes are contaminated. Injection is another way that chemicals enter the body. Though less common in most workplaces, it can occur when a sharp object (e.g., a needle) punctures the skin and injects a chemical (or virus) into the bloodstream. This can also occur when a chemical is sprayed at the body at high pressure. Eyes are also another route of entry, although usually only very small amounts of chemicals in the workplace enter through the eyes or mouth.

However the chemical enters the body, it is distributed throughout the body via the bloodstream, where they can attack and harm organs which are far away from the original point of entry as well as where they entered the body.

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Employer has to develop and carry out programmes spelling out methods and procedures on safety and health in the use of chemicals at workplace. A programme for the safe management of chemical should consist three(3) ideas;

- (i) Management should know all the chemicals being used in the enterprise, including their quantities and their associated hazards
- (ii) Workers should know the hazards associated with the chemicals they work with and should be trained in the necessary precautions.
- (iii) Workplaces should be designed or adapted to the needs of the worker instead of attempting to adapt the worker to the workplace.

Chemical control programme should include;

- a) Cooperation at the workplace

To ensure the role of employers and employees in handling chemicals at the workplace.

- b) Inventory of chemicals - Chemical Register

Employers has to identify and record all chemical hazardous to health processed, produced or stored in the place of work and maintained in good order and condition and be updated from time to time. The register shall be accessible to all employees at the place of work who may be exposed or are likely to be exposed to chemicals hazardous to health.

- c) Chemical Health Risk Assessment(CHRA)

Employers has to make sure that chemical health risk assessment is conducted in a workplace in order to identify, evaluate and control any health risk associated with work activities involving the use of chemicals. The purpose of conducting CHRA is to enabling decisions to be made on appropriate control measures, induction and training of employees, monitoring and health surveillance activities as may be required to protect the health of employees who may be exposed to chemicals hazardous to health at work.

- d) Purchasing procedure, Classification and Labelling

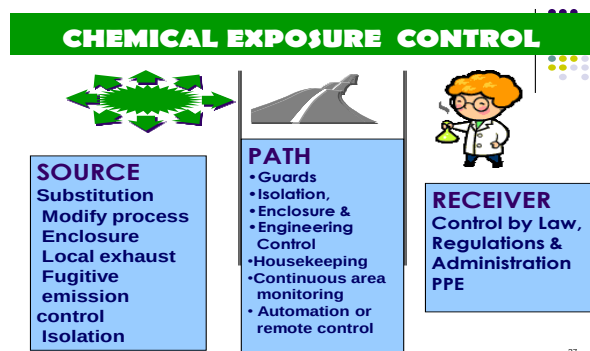
Each chemical entering the workplace must correctly identified, classified and labelled, complete with an up-to-date safety data sheet. Drums, bags and any other containers holding chemicals should be frequently checked for proper labels. The purpose of labelling is to warn workers of the potential

hazards of the chemicals, and necessary precautionary measures and what to do in case of an emergency. Label of chemical should consist of product Identifier, hazard statement, precautionary Statement, supplier identifications, hazard pictogram and signal word.

e) Safety Data Sheet (SDS)

Safety data sheet (SDS), intended to provide workers and emergency personnel with procedures for handling or working with a hazardous substance in a safe manner, and includes information such as physical data (melting point, boiling point, flash point, etc.), toxicity, health effects, first aid, reactivity, storage, disposal, protective equipment, and spill-handling procedures. It indicates appropriate precautions, including personal protective equipment as well as emergency procedures..

f) Ensure the adequacy of control measures



- Administrative Practices includes;
 - Time work so fewer workers are exposed,
 - Work upwind of mixing operations,
 - Shower after shift
 - Change clothes
 - No food or smoking in work areas
- Personal Protective Equipment(PPE) includes
 - Skin Protectionsuch as gloves, aprons, full body suits, and boots.
 - Respiratory Protection; respirators are effective against some chemicals but may provide little or no protection against others.
- Safe storage-elements includes

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- Separated from process areas
- Prevent mixing of incompatible e.g segregation, bunding
- Reduce risk of physical damage
- Prevent rapid spread of smoke/fire
- Minimise amount involved in incident -e.g necessary quantity, ventilation

g) Training and Education

People who work with the chemicals should be instructed and trained in the possible hazards caused by the chemicals, safe working procedures, the care and use of protective equipment, and emergency and first-aids measures.

h) Health Surveillance and Medical Removal Protection

Health surveillance to be conducted when CHRA indicates that health surveillance is necessary. Medical Removal Protection should be apply to an employee by remove him/her from any work that exposed or likely to expose him/her to chemicals hazardous to health.

i) Record keeping

Maintenance of record keeping by the employer on records related to chemical hazardous to health used at workplace should be in place.

j) Warning Sign

Employer to ensure on warning sign of chemical hazardous to health is posted to give warning to any person who may be in that area or who may be or likely to be risk of being affected by the chemicals hazardous to health.

k) Disposal

- Labelling of Wastes -Scheduled Waste Reg.
- Temporary Disposal Storage Area
- -weather protection & spill control
- Waste handlers
- -safe work procedure, use of PPE
- Safe disposal-DOE guidelines

5.2 Noise Control Management Programme

Employees should not be exposed to continuous noise levels exceeding 90dB(A) averaged over 8 working hours, or an 8-hour time-weighted average (TWA). The

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other two conditions are that no employee should exceed the 115 dB(A) at any time and no employees should exceed 140dB level at any time.

(i) Hearing Conservation Programme(HCP)

HCP is a set of carefully planned activities carried out with the aim of protecting and maintaining the hearing of employees. Aspect of HCP is based on requirements under the Noise Exposure Regulations 1989

- a) Conduct noise monitoring to monitor noise exposure levels in a way that accurately identifies employees exposed
- b) Control of noise or ppe - the recommended method of approach goes according to the order outlined below.
 - Plant planning (design and layout)
 - Substitution (equipment, process , material)
 - Engineering control
 - Hearing Protective Devices - Hearing protectors must adequately reduce the noise level for each employee's work environment.
- c) Established an audiometry test - monitors an employee's hearing over time. The important elements of the program include baseline audiograms, annual audiograms, training, and followup procedures.
- d) Training and education program - Employers must train employees in the effects of noise; the purpose, advantages, and disadvantages of various types of hearing protectors; the selection, fit, and care of protectors; and the purpose and procedures of audiometric testing
- e) Record keeping - Keeping the audiometric test records is important as the information is needed for medical surveillance. These records are needed to ensure employees are not overexposed and their hearing is not adversely affected during, before and after working hours.

5.3 Ergonomic Programme at the workplace

Ergonomics is the science of fitting machines, tools, and work processes to match human characteristics and capabilities. Potential Ergonomic Risk Factors -

Excessive force, awkward or non-neutral postures, static postures, repetitive motion, mechanical stress, prolonged sitting/standing, vibration, contact stress, cold temperatures, and poor lighting. Musculoskeletal disorders (MSDs) one of the disease related to ergonomics problem, affect the muscles, nerves and tendons. Work related MSDs (including those of the neck, upper extremities and low back). Workers in many different industries and occupations can be exposed to risk factors at work, such as lifting heavy items, bending, reaching overhead, pushing and pulling heavy loads, working in awkward body postures and performing the same or similar tasks repetitively. Exposure to these known risk factors for MSDs increases a worker's risk of injury

i) Ergonomics Programme

The purpose of the Ergonomics Program is to establish a system for the identification, evaluation, and control of ergonomic risk factors in the workplace. The goal of the ergonomics program is to maximize employee health, safety, efficiency, and productivity by matching the requirements of the job with the abilities of the worker. Measurable outcomes are:

- i. increased productivity
- ii. reduced worker compensation/medical costs
- iii. reduced absenteeism and job turnover
- iv. improved employee morale

Components of Ergonomic Programme include;

(a) Responsibilities

Management/Employer	Define clear goals and objectives for the ergonomic process, development, implementation, and management of the Ergonomics Program as needed and provide financial.
Employees - Workers	Identify and provide important information about hazards in their workplaces, assist in the ergonomic by reducing exposure to risk factors and evaluating the changes made as a result of an ergonomic assessment.

Supervisors	Participate in worksite analysis and ergonomic hazard analysis, refer employees complaints, train employees and utilize ergonomic principles when designing work practices and procedures.
Occupational Health Services	Provides prompt assessment, treatment, and follow up to employees reporting work-related musculoskeletal disorders.
Environment,Health, and Safety Committee (EHS)	Identifies potential ergonomic problems and contributing

(b) Program Elements

Worksite Analysis	Identifies and documents ergonomic risk factors which include - Walkthroughs of work areas; - Analysis of tool design, equipment, furniture, and workprocesses; - Measurement for proper adjustments; - Employee interview and surveys; - Injury and illness reports and trend analysis; - Documentation of existing or potential ergonomic hazards bymembers of EHS.
Prevention and Intervention Strategies	Depend on such factors as risks, costs of interventions, and presence ofsymptomatic individuals. The most effective intervention is the eliminationof the risk.Control strategies, in order of preference include engineering control, administrative control, health management and training and education.

(c) Record keeping

- Existing OHS, safety, worker compensation insurance records, all documents related are analyzed for evidence of injuries or disorders associated with cumulative trauma disorders.
- Ergonomic analyses of work areas identified by employees or management will be documented by employer.

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5.4 Confined Space Programme

Working in a confined space can expose a person to a very risky situation that can cause diseases and injury which leads to an accident and death. Statistics have shown many fatalities involving workers working in a confined space such as working in a silo, storage tank, pressure vessel or tank, sewer system or communication system, trenches and others. In most cases the cause of death is due to inhaling of toxic gases or oxygen deficiency.

i) Confined space entry programme

The confined space entry programme shall include at least the following elements

- a) **Hazard identification** - Identify all hazards associated with working in the confined space.
- b) **Risk assessment** - Carry out the risk assessment before carrying out work involving entry into a confined space
- c) **Risk control** - Based on the risk assessment, establish and implement the procedures and practices of hazard control that provide safe entry into the confined space.
- d) **Stand-by person** - At least one stand-by person outside the confined space into which entry is authorised for the duration of entry operations.
- e) **Monitoring of multiple confined spaces** - For monitoring of multiple confined spaces by a single stand-by person, the programme shall include the means and procedures to enable the stand-by person to respond to an emergency affecting one or more of the confined spaces being monitored without distraction from the stand-by person's responsibilities
- f) **Permit system** - Develop a written system for preparation, issuance, implementation, and cancellation of permit to work.
- g) **Sign-posting** - Confined space warning signages to be posted at conspicuous places when work is being carried out after the issuance of the permit to work.
- h) **Employee training** - Ensure that employees who are involved in confined space operation have undergone training on confined space safety course approved by the Director General of DOSH and passed the examination.

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- i) **Equipment** -Provide appropriate equipment use for testing and monitoring; personal protective emergency and rescue equipment; and any other equipment necessary for safe entry into and rescue from confined spaces
- j) **Means of Communication**—such as telephone or walkie talkie
- k) **External hazard protection**- Protection such as physical barriers to be provided to control potential hazards posed by pedestrians and vehicles.
- l) **Identification of the duties** - identify the duties of employees to be an authorised entrant, stand-by person, entry supervisor, authorised gas tester, permit issuer or other related personnel in entry operation.
- m) **Information to the contractor** - brief contractors with information about the confined space hazards, safety rules and emergency procedures
- n) Coordination of entry operations involving employees of more than one contractor working simultaneously
- o) Conclusion of entry

SOALAN/QUESTION :

1. What are five factors affecting the successful implementation of a safety and health programme ?
2. Name three conditions that must exist for the successful of a safety and health programme ?
3. What are the objectives safety and health programme ?
4. List five benefits of implementation of safety and health programme ?
5. What is the purpose of worksite analysis ?
6. What is tool box talk ?
7. What is lock-out and tag-out system ?
8. What types of work permit used in work activities ?
9. List the three ideas that safe chemical management programme should consist.
10. State three conditions of noise exposure levels that no employee should exceed and the legislation related

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1. Factor affecting the **successful implementation** of a safety and health programme are :
 - Management commitment on the allocation of monetary budget and the demonstration of job safety and health;
 - Training of supervisor and new employees;
 - An established written safety and health programme;
 - Accountability for safety and health at all levels;
 - Response and compliance to accident investigation; and
 - Conduct safety and health inspection and audit

2. There are three condition which must exist for a successful safety and health programme :
 - Management leader ship
 - Safe Working Condition
 - Safe working habits by all employees

4. The objectives of a safety and health programme are to ensure ;
 - The company has a direction and commitment for safety and health that include policies and procedures for all employees and other interested parties.
 - All misunderstanding and misconceptions are clarified for a specific safety and health activity or issue;
 - The practise of safety and health at the workplace is part of the objectives the performance measurement;
 - There is a written safety and health rules and procedure for every to follow;
 - That all activities have been integrated with safety and health requirement;
 - Proper communication tools are established for safety and health communication

5. To reduce unwanted incident or accidents and also damage to property. Other benefit include :
 - Reduces company`s insurance premium
 - Reduces indirect cost of accidents
 - Prevent potential negative impact on company image or repotation

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- Reduces physiological effects on employees due to accident
- Reduces long term or permanent disability cases
- Improves employee morale and loyalty
- Increases productivity
- Makes employees proud to be part of the company.

6. Worksite analysis is an activity that should be included in the safety and health programme. The purpose of this activity is to identify hazard and potential hazards that may exist in work processes and in the working environment. This activity is conducted through observation and by specially – trained personnel.

The following are performed to identify workplace hazards or potential hazards :

- Conduct comprehensive periodical surveys on hazards identification
- Analyse new processes, methods, materials, equipment and installation
- Conduct Hazard Identification, Risk Assessment and Risk Control (HIRARC) or job hazards analysis
- Conduct regular site safety and health inspection
- Provide a system that is reprisal –free for employees that notify management of hazardous conditions

7. A tool box talk is a hands on way to remind workers that safety and health are important on the job. It deal with specific problems on site. They do not replace formal training. Through tool box talks can tell workers about safety and health requirements for the tool, equipment, material and procedures they use every day or for particular jobs. Each talk will take about five minutes to present. In delivering tool box talks, the objective is to help workers recognize and control hazards on the project. Tool box demonstrate the commitment of employers and workers to safety and health on the job.

8. A Lock-out tag-out system is a written practice which when followed, will allow workers to carry out their job function without undue risks. The system's procedure can be quite simple, yet precise, such as locking out and tagging out simple piece of machinery or complex, such as the locking out of entire process. A lock-out tag out may applied to electrical and mechanical equipment in nature,

9. Types Of Permit Used in Work Activities

- Welding / burning (naked flame) work permit

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- Hot work (no naked flame) work permit.
- Flammable release work permit.
- Vessel entry work permit (confined spaces)
- Cold work permit.
- Electrical work permit.
- Safe excavation work permit
- Working at height work permit

10. A programme for the safe management of chemical should consist three(3) ideas,

- I. Management should know all the chemicals being used in the enterprise, including their quantities and their associated hazards
- II. Workers should know the hazards associated with the chemicals they work with and should be trained in the necessary precautions.
- III. Workplaces should be designed or adapted to the needs of the worker instead of attempting to adapt the worker to the workplace

11. Under Factories and Machinery (Noise Exposure Regulations) 1989, Employees should not be exposed to ;

- I. continuous noise levels exceeding 90dB(A) averaged over 8 working hours, or an 8-hour time-weighted average (TWA).
- II. Exceed 115 dB(A) at any time
- III. Exceed 140dB level at any time

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